x_1	x_2
-2	-1
-1	0
0	1
1	2
2	3

$$u_1 = \frac{-2+(-1)+0+1+2}{5} = 0$$

$$u_2 = \frac{-1+0+1+2+3}{5} = 1$$

$$u = [0\ 1]$$

$$X_c = \begin{bmatrix} -2 & -2 \\ -1 & -1 \\ 0 & 0 \\ 1 & 1 \\ 2 & 2 \end{bmatrix}$$

$$C = \frac{1}{n-1} X_c^T X_c$$

$$= \frac{1}{4} \begin{bmatrix} -2 & -1 & 0 & 1 & 2 \\ -2 & -1 & 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} -2 & -2 \\ -1 & -1 \\ 0 & 0 \\ 1 & 1 \\ 2 & 2 \end{bmatrix}$$

$$= \frac{1}{4} \begin{bmatrix} 10 & 10 \\ 10 & 10 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{5}{2} & \frac{5}{2} \\ \frac{5}{2} & \frac{5}{2} \end{bmatrix}$$

$$\begin{vmatrix} \frac{5}{2}-\lambda & \frac{5}{2} \\ \frac{5}{2} & \frac{5}{2}-\lambda \end{vmatrix} = 0$$

$$\left(\frac{5}{2}-\lambda\right)^2 - \left(\frac{5}{2}\right)^2 = 0$$

$$\frac{25}{4} - 5\lambda + \lambda^2 - \frac{25}{4} = 0$$

$$\lambda(\lambda - 5) = 0$$

$$\lambda_1 = 5 \quad \lambda_2 = 0$$

$$V_1 = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad V_2 = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$(C - 5I)w = 0$$

$$\begin{bmatrix} -\frac{5}{2} & \frac{5}{2} \\ \frac{5}{2} & -\frac{5}{2} \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} = 0$$

$$\begin{bmatrix} -\frac{5}{2}w_1 + \frac{5}{2}w_2 \\ \frac{5}{2}w_1 - \frac{5}{2}w_2 \end{bmatrix} = 0$$

$$\|w\|_2 = \sqrt{w_1 + w_2} = 1$$

$$2w_1^2 = 1$$

$$w_1 = w_2 = \frac{1}{\sqrt{2}}$$

$$wPC_1 = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$x = \begin{bmatrix} 3 \\ 3 \end{bmatrix} \quad u = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$x_c = x - u = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$

$$z = w^T x_c$$

$$z = \frac{1}{\sqrt{2}}[1\ 1]\begin{bmatrix} 3 \\ 2 \end{bmatrix}$$

$$z = \frac{5}{\sqrt{2}}$$

$$\hat{x} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} + \frac{5}{\sqrt{2}}\frac{1}{\sqrt{2}}\begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad / \quad \hat{x} = u + zw$$

$$= \begin{bmatrix} 0 \\ 1 \end{bmatrix} + \frac{5}{2}\begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$= \frac{1}{2}\begin{bmatrix} 5 \\ 7 \end{bmatrix}$$

$$Error = \|x - \hat{x}\|^2$$

$$= \sqrt{\left(3-\frac{5}{2}\right)^2 + \left(3-\frac{7}{2}\right)^2}$$

$$= \frac{1}{\sqrt{2}}$$

$$x = \begin{bmatrix} 3 \\ 4 \end{bmatrix} \quad u = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$x_c = \begin{bmatrix} 3 \\ 3 \end{bmatrix}$$

$$z = w^T x_c$$

$$= \frac{1}{\sqrt{2}}[1\ 1]\begin{bmatrix} 3 \\ 3 \end{bmatrix}$$

$$= \frac{6}{\sqrt{2}}$$

$$\hat{x} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} + \frac{6}{\sqrt{2}}\frac{1}{\sqrt{2}}\begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

$$Error = \|x - \hat{x}\|^2$$

$$= 0$$